

Training a Minimal Transformer for 10-Digit Addition

Codex Experiment Report

Autonomous run in `addition_experiment_codex_ver2`

February 19, 2026

Executive Summary

Objective: train the smallest decoder-only transformer that reaches at least 99% exact-match accuracy on held-out 10-digit addition, with standard autoregressive generation and cross-entropy training.

Final selected model:

- 1 layer, $d_{model} = 8$, 2 heads, $d_{ff} = 12$
- Context length: 23 tokens
- Vocabulary size: 114 tokens
- Total parameters: **1,644**

Held-out test result (10,000 examples): **99.04% exact-match** (0.9904). This meets the target and is the smallest passing configuration found in this run.

1 Task Setup and Constraints

The task required deterministic preprocessing and postprocessing, no symbolic solver at inference time, and a minimum 10,000-example held-out random test set. The model had to generate autoregressively and was trained with cross-entropy. All artifacts for this experiment were kept inside one folder.

2 Data Pipeline

2.1 Tokenization and Formatting

The successful format used a **pair-column tokenization**:

- Output digit tokens: 0..9 (10 tokens)
- Input pair tokens: P00..P99 (100 tokens)
- Specials: =, <bos>, <eos>, <pad> (4 tokens)

Total vocabulary size: $10 + 100 + 4 = 114$.

For operands A, B (both zero-padded to 10 digits), define reversed digits $a_0..a_9, b_0..b_9$ from least-significant to most-significant.

$$\text{preprocess}(A, B) = [\text{<bos>}, P(a_0, b_0), P(a_1, b_1), \dots, P(a_9, b_9), =]$$

Target sequence:

$$[c_0, c_1, \dots, c_{10}, \text{<eos>}], \quad c_i \text{ are digits of } (A + B) \text{ in reversed order.}$$

2.2 Deterministic Postprocess

Given generated tokens, read digits until `<eos>` or non-digit, keep/pad to 11 digits, reverse back, and parse as integer. This implements deterministic `postprocess(model_output)`.

2.3 Split Hygiene

Validation and test sets were generated once from fixed seeds and saved. During training, sampled (A, B) pairs were explicitly checked against reserved holdout hashes and resampled on collision.

3 Model Architecture

Decoder-only transformer with pre-norm blocks, learned token/position embeddings, causal self-attention, GELU MLP, and tied input/output embeddings.

Final best checkpoint architecture:

Component	Value
Layers	1
Hidden dim d_{model}	8
Heads	2
Feedforward dim d_{ff}	12
Dropout	0.0
Context length	23
Vocabulary size	114
Total parameters	1,644

4 Training Configuration

- Optimizer: AdamW
- LR schedule: linear warmup + cosine decay
- Final run hyperparameters: $\text{LR} = 3 \times 10^{-3}$, warmup 100, min LR ratio 0.1, weight decay 0.01, grad clip 1.0
- Batch size: 512
- Steps (final run): 5,000
- Examples seen (final run): 2,560,000

- Validation: every 300 steps on 2,000 held-out examples
- Hardware: CPU
- Final run wall time: ~ 95.4 seconds

Loss masking: cross-entropy was applied only to answer tokens (11 digits + `<eos>`), not to prompt tokens.

5 Experiment Process and Iteration Log

The work followed a staged process:

1. Build full training/eval pipeline and run smoke checks.
2. Detect failure of initial raw-digit format under tiny models.
3. Redesign tokenization to pair tokens `Pab` and verify viability.
4. Perform aggressive architecture search toward minimum parameter count.
5. Validate best candidate on a 10,000-example held-out test set.

5.1 Run Outcomes

Table below lists all tracked runs with best validation exact-match.

Run	Params	Best val EM	Best step
search_l1_d6	1188	0.0000	0
search_l1_d7_h1	1442	0.1910	3000
search_l1_d8_ff8	1576	0.6875	3900
search_l1_d8_ff8_tuned	1576	0.0000	0
search_l1_d8_ff10	1610	0.1705	3150
search_l1_d8_ff11	1627	0.7335	3900
search_l1_d8_ff12	1644	0.9935	4999
search_l1_d8	1712	0.9845	2999
search_l1_d8_long	1712	1.0000	1800
pilot_l2_d32	18720	0.0000	0
smoke	18720	0.0000	0
pilot_pair_l2_d32	21536	1.0000	800
smoke_pair	21536	0.0000	0
baseline_l3_d64	103680	0.0000	0

Key findings from search:

- `search_l1_d8_ff12` (1,644 params) crossed the target (best val EM 0.9935).
- Smaller nearby models (e.g., 1,627 and 1,610 params) did not approach 99% in observed runs.
- 1,712-parameter variant reached 100% validation but was not minimal.

6 Training Curves and Diagnostics

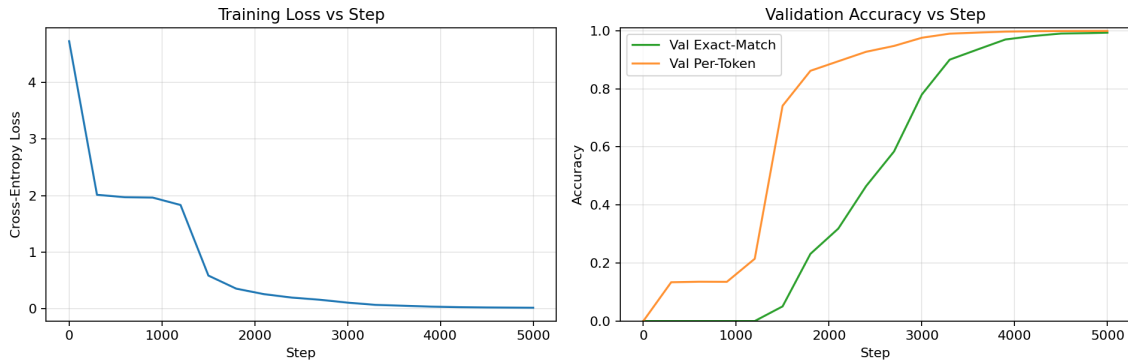


Figure 1: Final selected run: training loss and validation accuracy vs. step.

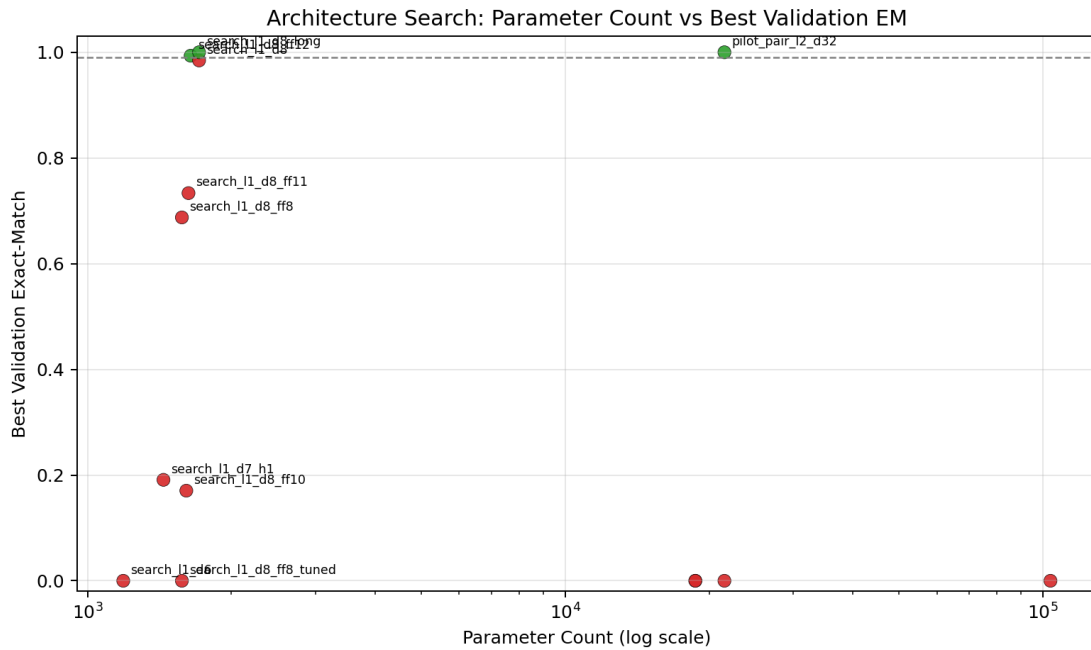


Figure 2: Architecture sweep summary: parameter count vs. best validation exact-match.

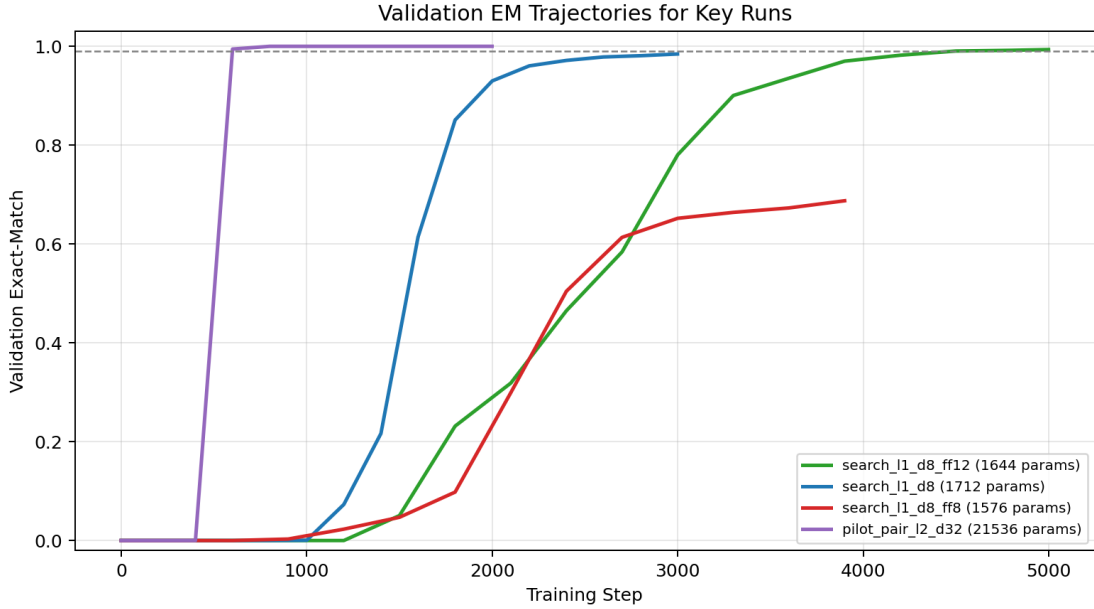


Figure 3: Validation exact-match trajectories for representative runs.

7 Final Evaluation (Held-Out Test, 10,000 Examples)

Checkpoint: `artifacts/search_l1_d8_ff12/checkpoints/best.pt`

Metric	Value
Exact-match accuracy	0.9904
Token-level digit accuracy	0.9990727
Test set size	10,000

Since exact-match is below 100%, failure samples are included (10 examples):

1. A=5494929602, B=5097590747, prediction=10592510349, truth=10592520349
2. A=2387190696, B=8725060082, prediction=11112260778, truth=11112250778
3. A=9930973050, B=9755128054, prediction=9686101104, truth=19686101104
4. A=8260033151, B=2937091401, prediction=11197024552, truth=11197124552
5. A=9310394203, B=9711475589, prediction=9021869792, truth=19021869792
6. A=8493050971, B=2211750872, prediction=10704802843, truth=10704801843
7. A=6999720508, B=4969121632, prediction=11969842140, truth=11968842140
8. A=6809322292, B=1649563090, prediction=8459885382, truth=8458885382
9. A=9466379984, B=894469793, prediction=10360840777, truth=10360849777
10. A=822370053, B=9449501863, prediction=10271881916, truth=10271871916

8 What Worked Best

- Reversed output aligned generation order with carry propagation.
- Pair-token input collapsed each digit-column into one token, reducing alignment burden.
- Strict holdout exclusion during sampling ensured clean generalization measurement.
- Compact architecture search around the capacity boundary identified the minimum passing model.

9 Reproducibility

All artifacts are in this folder tree. Key files:

- Training/eval pipeline: `experiment.py`
- Final run config: `artifacts/search_l1_d8_ff12/config.json`
- Final run metrics: `artifacts/search_l1_d8_ff12/metrics.csv`
- Final run summary: `artifacts/search_l1_d8_ff12/summary.json`
- Final test results: `artifacts/search_l1_d8_ff12/test_results.json`

Primary commands used:

```
python3 experiment.py train --run-name search_l1_d8_ff12 ...
python3 experiment.py eval --ckpt artifacts/search_l1_d8_ff12/checkpoints/best.pt ...
python3 experiment.py plot --metrics-csv artifacts/search_l1_d8_ff12/metrics.csv \
  --out-png artifacts/search_l1_d8_ff12/curves.png
```